THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** metre	SI unit of length is metre (m).
2	**(d)** material	An object is made of material.
3	**(b)** temperature	Temperature measures hotness.
4	**(a)** three	Solid, liquid and gas.
5	**(a)** thread	A thread follows the curve.
6	**(c)** sugar	Sugar dissolves.
7	**(a)** mercury	Mercury expands uniformly.
8	**(a)** temperature is higher	Heat helps.
9	**(a)** changes with time	Change of position.
10	**(a)** copper	Metals conduct heat well.
11	**(a)** conductors	Metals are good conductors.
12	**(d)** collecting and storing rainwater	Collect rain for use.
13	**(d)** 100 centimetres	1 m = 100 cm.
14	**(d)** transparent	We can see clearly through them.
15	**(c)** thermometer	Thermometer measures temperature.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Everyone gets the same value (1); Results can be compared and shared (1).	1+1
17	The substance an object is made of (1); Wood and iron (1).	1+1
18	A measure of how hot or cold a body is (1); It is measured with a thermometer (1).	1+1
19	Solid: ice (1); Liquid: water; gas: water vapour (1).	1+1
20	Lay a thread along the path (1); Straighten the thread and measure it (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Place the zero of the ruler at one end (1); Keep your eye above the other end (1); Read the mark and write with unit (1).	1+1+1
22	Clear glass: transparent (1); Butter paper: translucent (1); Wooden door: opaque (1).	1+1+1
23	Wash and shake it below 35 °C (1); Place under the tongue or armpit for a minute (1); Read with the eye level to the mercury thread (1).	1+1+1
24	Sun heats water bodies (1); Vapour rises and cools (1); It condenses into tiny droplets forming clouds (1).	1+1+1
	OR: Zero may be missing (1); It gives wrong length (1); Use a clear full mark (1).	1+1+1
25	Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred fast (1).	1+1+1
	OR: Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred to the hand (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Linear: a car, a falling stone (1); Circular: stone whirled on a string, hands of a clock (1); Rotational: top, fan (1); Periodic: pendulum, swing (1); Random: butterfly (1).	1+1+1+1+1
	OR: Appearance: lustre of gold (1); Hardness: diamond (1); Solubility: sugar (1); Transparency: glass (1); Floating: cork, sinking: stone (1).	1+1+1+1+1

27	Glass tube with mercury bulb (1); Mercury expands on heating (1); Kink keeps the level (1); Shake before use (1); Do not hold by the bulb (1).	1+1+1+1+1
	OR: Sun heats sea (1); Evaporation (1); Condensation to clouds (1); Precipitation (1); Water flows back to rivers and seas (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** water vapour	Vapour is invisible water gas.
2	**(a)** directly above the mark	Avoids parallax error.
3	**(a)** sand	Sand does not dissolve.
4	**(d)** prevents mercury from falling back on its own	It keeps the reading.
5	**(b)** the surface area is larger	More surface is exposed.
6	**(a)** the seat of the train	Relative to the seat, no change.
7	**(b)** poor conductors of heat	They do not become hot.
8	**(d)** insulators	Wood, plastic, air.
9	**(a)** closing taps when not in use	Prevent wastage.
10	**(b)** 1000 metres	1 km = 1000 m.
11	**(d)** opaque	Wood and metal are opaque.
12	**(d)** degree Celsius	Celsius scale.
13	**(a)** evaporation	Evaporation occurs at all temperatures.
14	**(d)** parallax error	Reading from an angle gives parallax.
15	**(b)** wooden block	Wood is less dense than water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They are poor conductors of heat (1); So hands do not get burnt (1).	1+1
17	Its range is limited (1); It would break (1).	1+1
18	Continuous movement of water on the Earth (1); Evaporation, condensation and precipitation (1).	1+1
19	They differ from person to person (1); Results will not agree (1).	1+1
20	It is transparent (1); We can see through it (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ice melts to water at 0 °C (1); Water heats up to 100 °C (1); It boils to steam (1).	1+1+1
22	Place zero mark at the start (1); Read from above (1); Use a full mark if the end is broken (1).	1+1+1
23	Hard materials are not easily scratched (1); Used in tools, buildings (1); Soft materials used for cushions (1).	1+1+1
24	Reduce heat exchange (1); Keeps food hot or cold (1); Helps in travel (1).	1+1+1
	OR: Cooling of air (1); Condensation of vapour (1); Droplets near ground (1).	1+1+1
25	4 km = 4000 m (1); Add 250 m (1); Total = 4250 m (1).	1+1+1
	OR: Take a glass of water (1); Add a spoon of the substance and stir (1); If it disappears, it is soluble (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Measure 100 sheets together (1); Total thickness is 1.2 cm (1); Divide by 100 (1); Gives 0.012 cm (1); Convert to 0.12 mm (1).	1+1+1+1+1
	OR: Place objects in water (1); Wood, cork float (1); Stone, nails sink (1); Float if lighter than water (1); Record observation (1).	1+1+1+1+1
27	One hand in hot water, one in cold (1); Then both in lukewarm water (1); Hot hand feels cold (1); Cold hand feels warm (1); Therefore use a thermometer (1).	1+1+1+1+1
	OR: Drinking, cooking (1); Farming (1); Close taps (1); Fix leaks (1); Harvest rain (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** shaken to bring mercury below 35 °C	Reset the mercury level.
2	**(a)** wind carries away water vapour	Wind speeds evaporation.
3	**(d)** the person is moving	Rest and motion are relative.
4	**(a)** wool	Wool comes from sheep.
5	**(d)** aluminium	Metals conduct.
6	**(b)** flood	Flooding.
7	**(d)** 10 millimetres	1 cm = 10 mm.
8	**(b)** translucent	Frosted glass is translucent.
9	**(d)** it can be misleading	Past experience affects touch.
10	**(c)** condensation	Cooling of vapour.
11	**(b)** start measuring from a full mark like 1 cm and subtract it	Use a clear mark and subtract.
12	**(d)** iron nail	Iron is denser.
13	**(b)** bulb	Bulb ought to touch the body only.
14	**(c)** air already has much water vapour	High humidity.
15	**(a)** rotational	Rotation about its axis.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Metals are good conductors of heat (1); Food is cooked quickly (1).	1+1
17	Heat speeds evaporation (1); Wind carries vapour away (1).	1+1
18	1 m = 100 cm (1); 2.5 x 100 = 250 cm (1).	1+1
19	Transparent: light passes through (1); Opaque: light is blocked (1).	1+1
20	It can be misleading (1); Warm hand feels lukewarm water as cold (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Keep the bulb in the liquid (1); Do not touch the vessel base (1); Read with the eye level (1).	1+1+1
22	Evaporation from water bodies (1); Condensation into clouds (1); Precipitation as rain or snow (1).	1+1+1
23	Different lengths for different people (1); Not accurate (1); Cannot be compared (1).	1+1+1
24	Heat one end of a rod (1); Place a wax-held pin on the other end (1); If pin falls quickly, it is a good conductor (1).	1+1+1
	OR: Heat flows from hot to cold (1); Mixture has an intermediate temperature (1); Both finally have the same temperature (1).	1+1+1
25	Water evaporates (1); It goes into the air (1); Faster in heat (1).	1+1+1
	OR: Body parts used (1); Cubit, span (1); Standard units made later (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Place thread on the curve (1); Mark the end (1); Straighten and measure (1); Wind thread once round bottle (1); Measure the thread (1).	1+1+1+1+1
	OR: Collect objects (1); Hold each in front of a torch (1); Observe the light passing (1); Group as transparent, translucent, opaque (1); Record in a table (1).	1+1+1+1+1

27	Take liquid in a beaker (1); Dip the bulb without touching the base (1); Wait for the reading to stay steady (1); Read with the eye level (1); Write with the unit (1).	1+1+1+1+1
	OR: Solid: ice (1); Liquid: water (1); Gas: vapour (1); Melting, freezing (1); Evaporation, condensation (1).	1+1+1+1+1

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